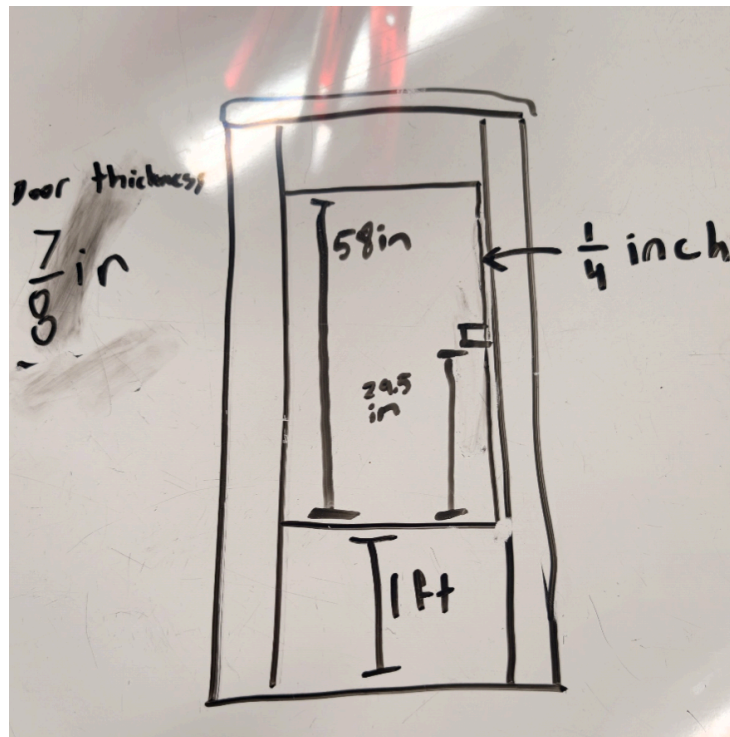


Context:

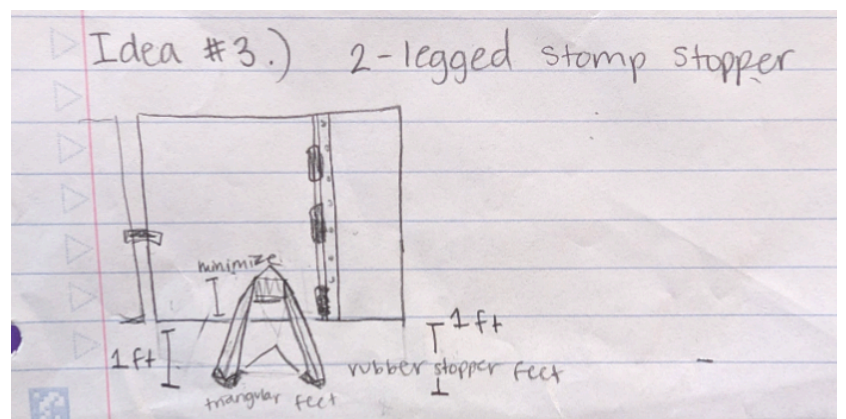
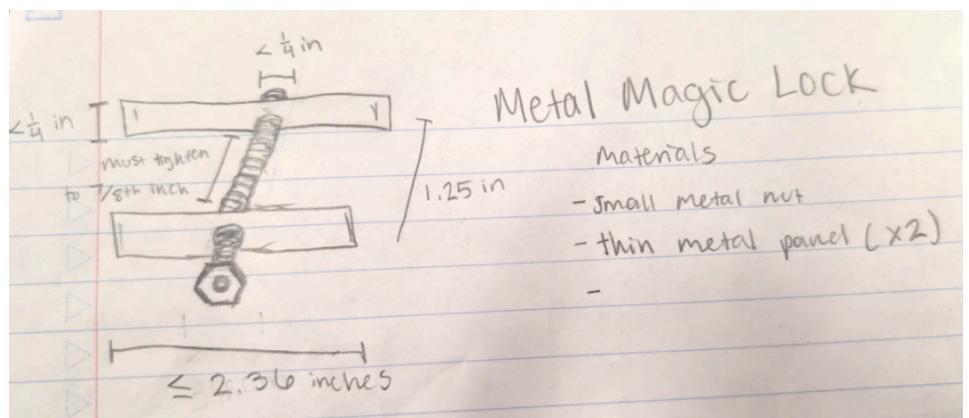
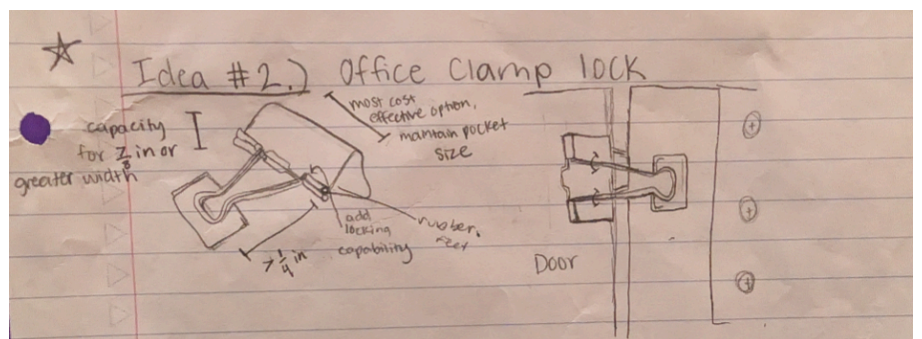
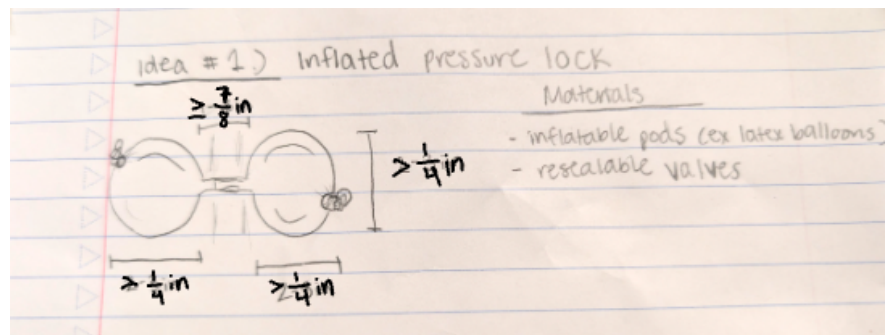
Door measurements (stem bathroom) to represent general stall door style:



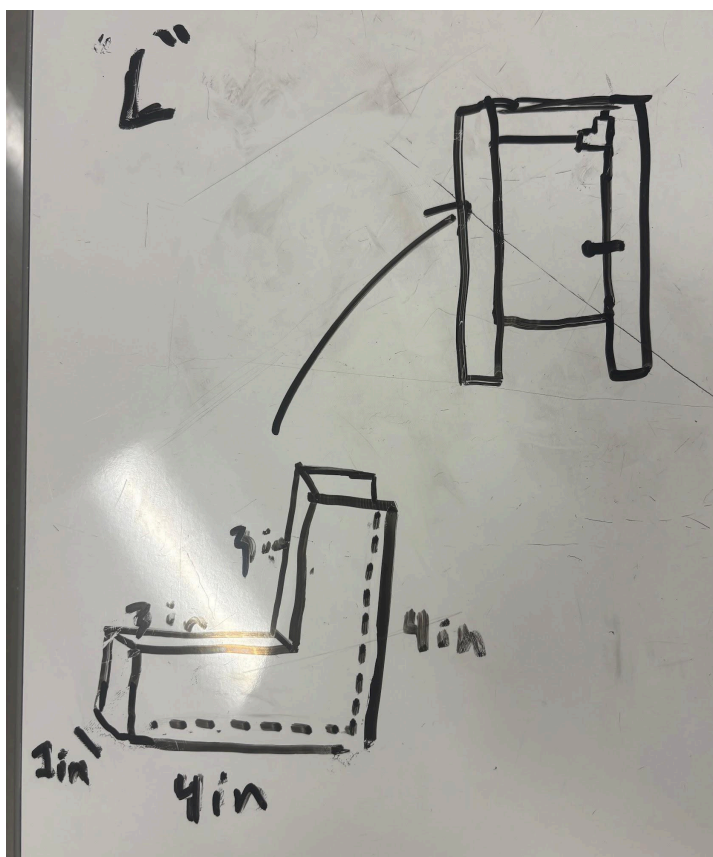
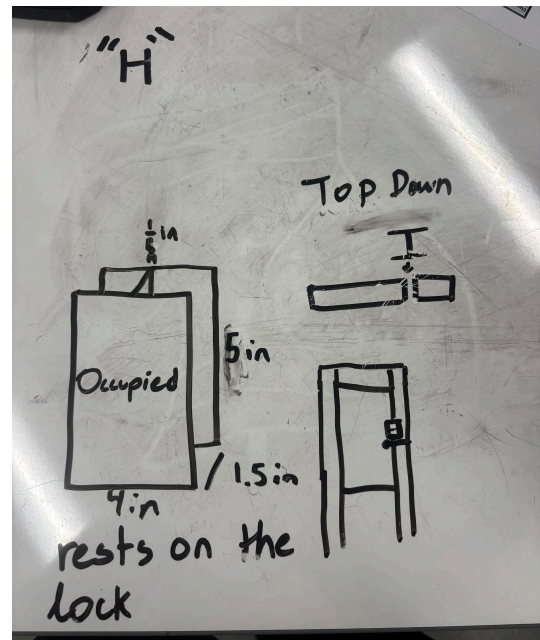
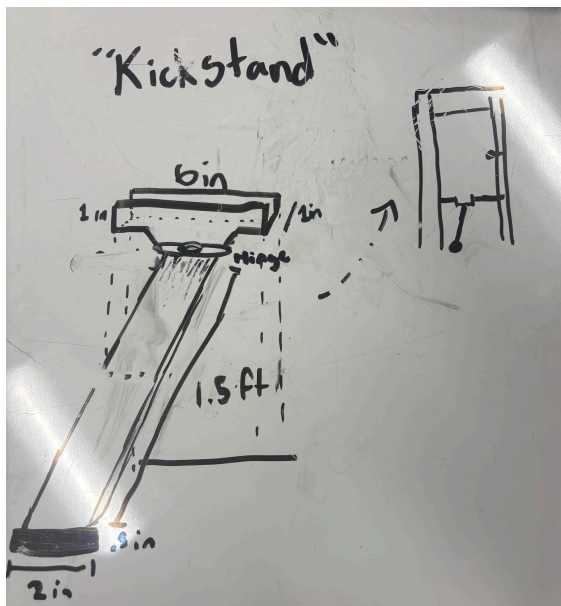
*Note: all prototype drawings are drawn with scales aligning with the above measurements to ensure their utility.

Individual Member Ideas:

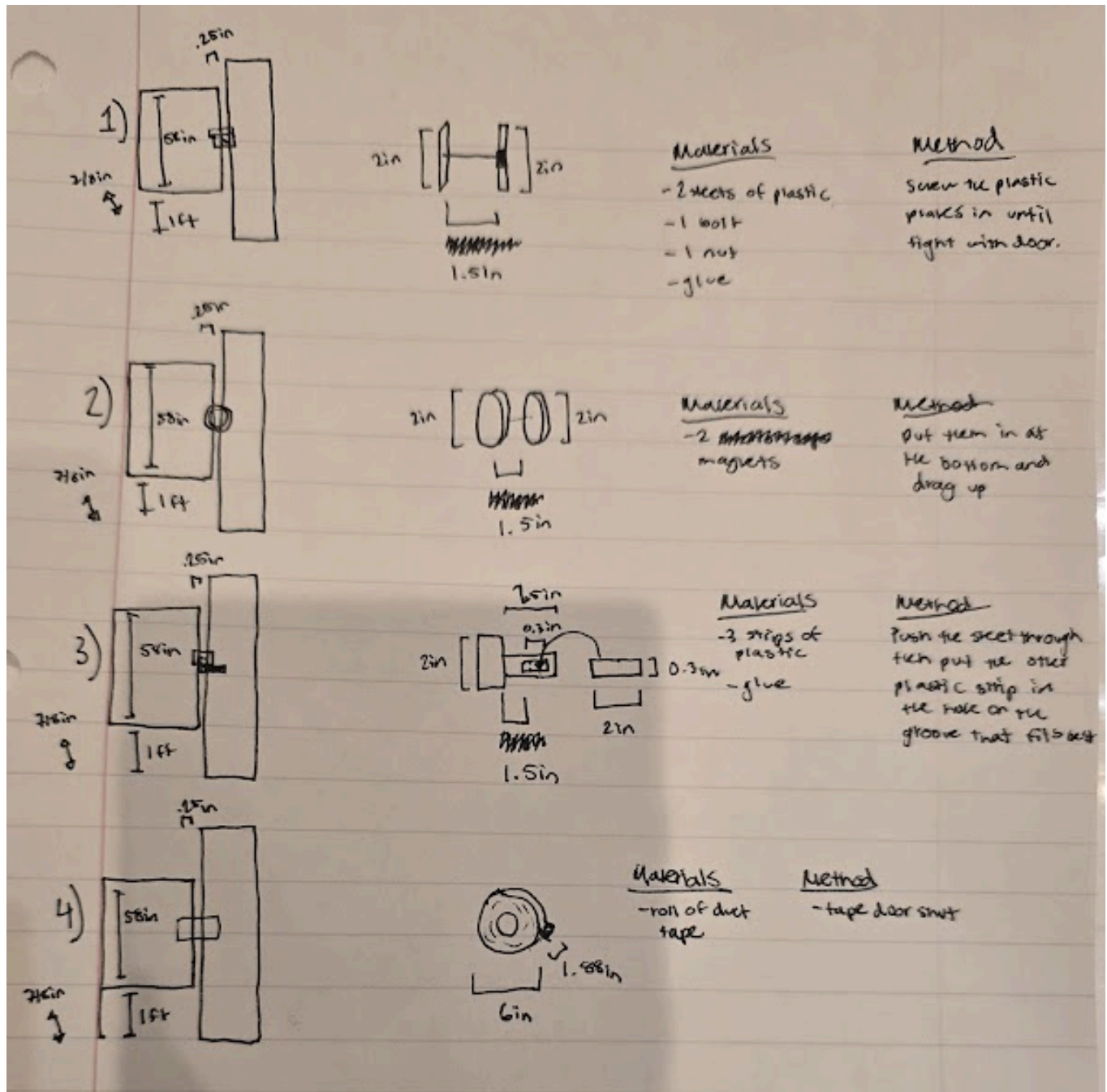
Kayla:



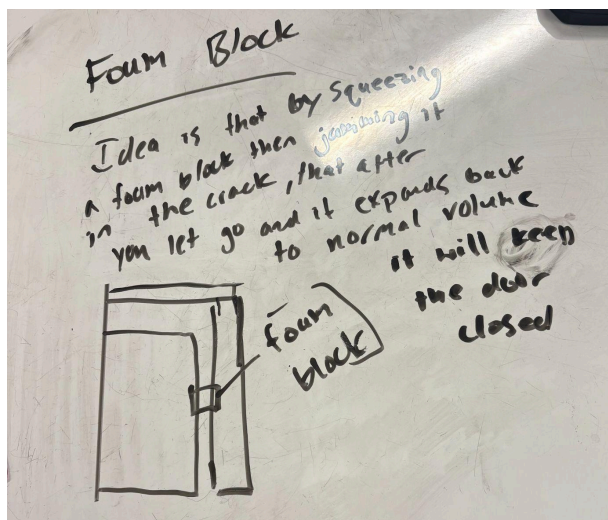
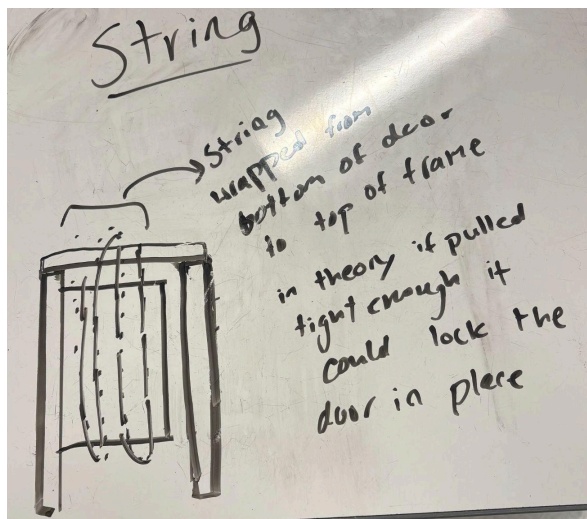
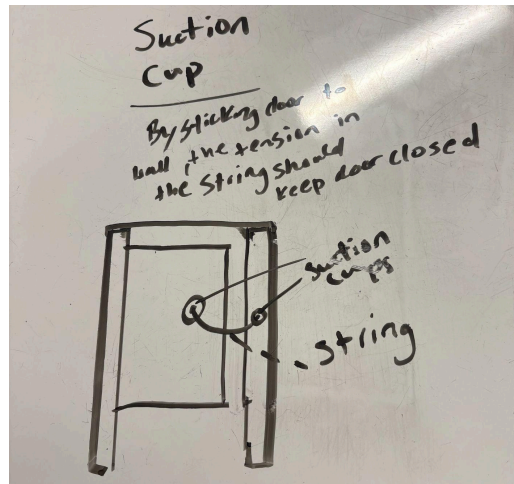
Keenan:



Meredith:



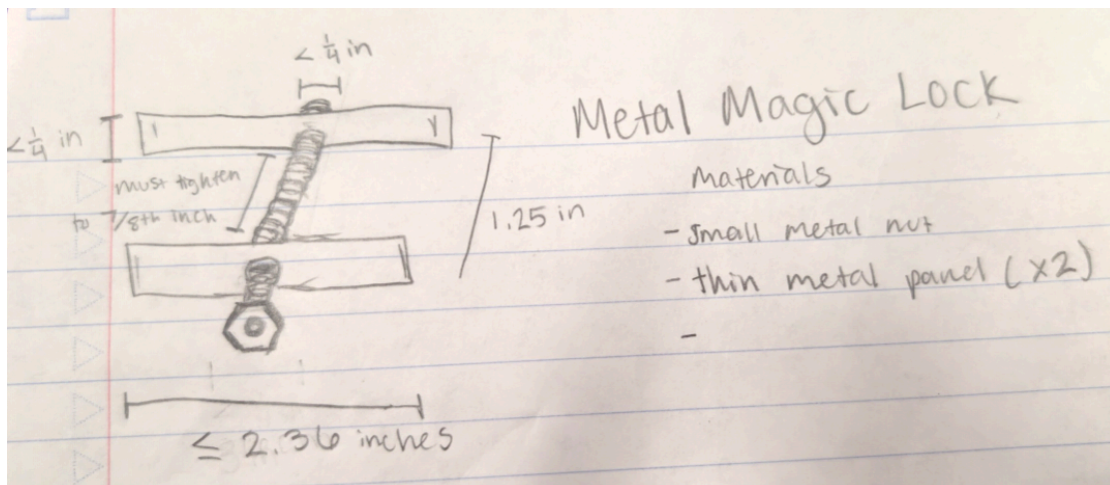
Jack:



Top Design from each Member:

We had each member individually decide on their best idea to present to the group for group decision making. Listed below are each group members' self-rated best ideas;

Kayla:

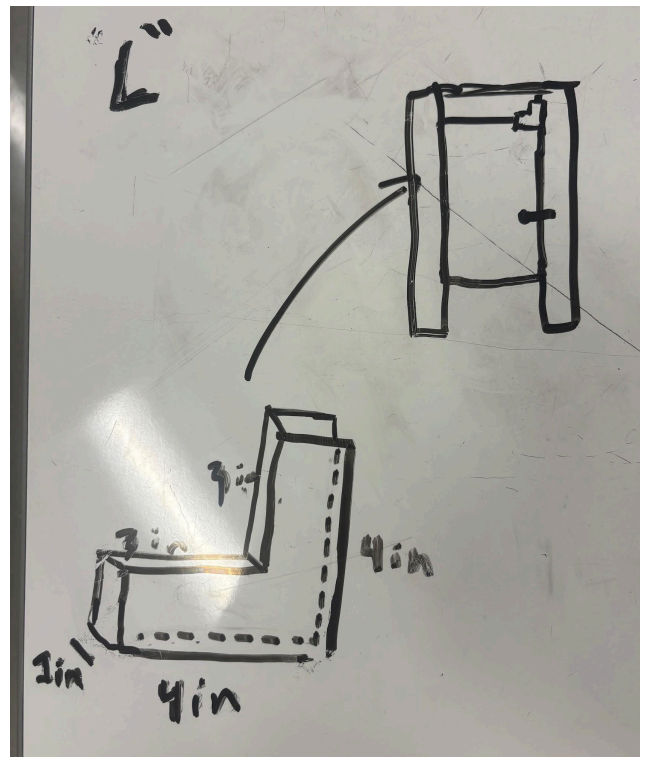
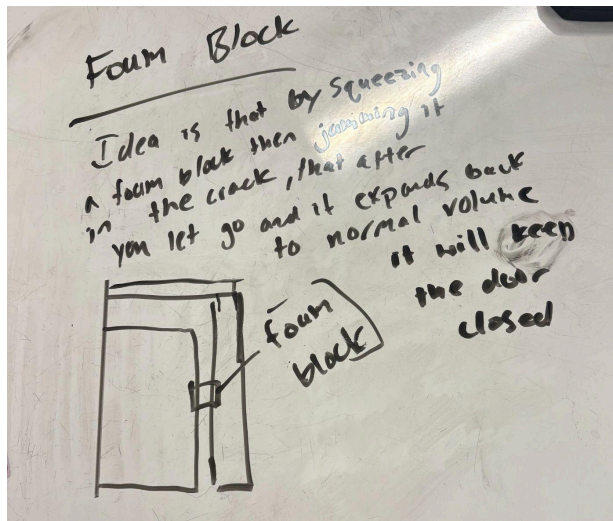


Why the Metal Magic Lock?

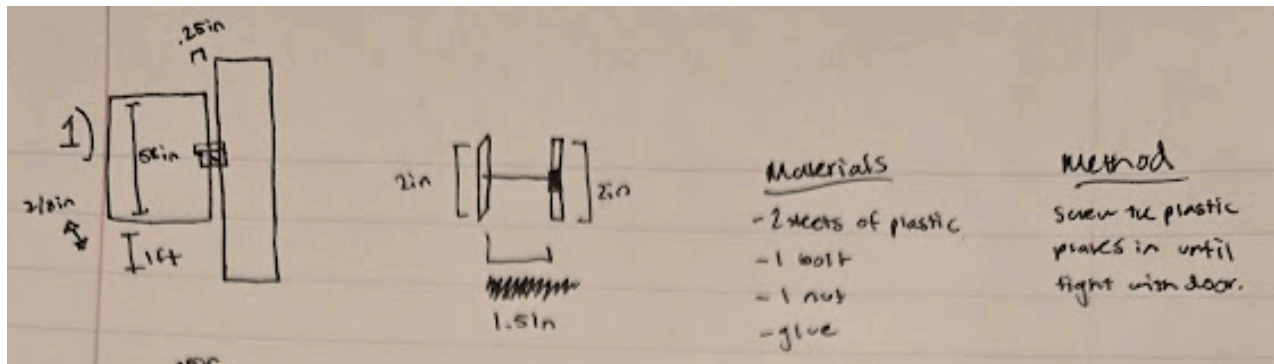
I chose this prototype as my top design due to its adherence to the most essential criteria of our project, simplistic accessible design, and durability and portability. This lock is washable, waterproof, and practical, making it a more desirable option. My inflatable design has the potential to be difficult to execute, as it requires a more niche set of materials, and the stomp stopper lock might be too large to fit usefully in a typical travel bag.

Keenan:

I chose this design because of the simplicity of the design. I believed that this design would work on all door types. Even if the door was attached to a side piece that was the same height as the door, the bottom half of the L design would account for this. It was also the most ergonomic design choice that I had. I also believed that out of my three designs this would be easiest to construct while maintaining structural integrity.



Meredith:



I chose this design over my others because it seemed to require the least costly materials as you can buy these in individual supply as opposed to some of my other ideas which included magnets. I also thought that duct tape would be a worse idea as you could feasibly run out if you went to the restroom too much or forgot to buy a new roll before your next trip.

Additionally, I feared the adhesive would damage the bathroom stalls in certain instances where the stalls had been painted. For my 3rd idea, I ruled it out as it required the plastic to fit into a small groove which may have degraded over time, and the specific grooves may have been too far depending on the door needing to be used.

Jack:

I chose my foam block lock method because I believed it was the simplest solution. While the string idea was interesting, it definitely would have taken too much time to wrap around the door's structure. Upon further thinking on the suction cup, I am not sure how well the suction would continue to work since interaction with dirt and dust lessens the effectiveness of suction cups, and having to clean after every use would add too much work to the stakeholders--while the foam block isn't affected by dirtiness or time constraints. You take the foam block from your pocket, squeeze it and place it in the door crack, release it, and then in theory after the foam expands it would keep the door from swinging anymore. It fit the criteria well for weight/volume, cheap cost, and ability to bring through TSA.

Decision Matrix (based on criterion from part c)

The designs that Kayla and Meredith presented as their best designs were incredibly similar, so the final design choice for the two of them will be ranked the same in the decision matrix.

We ranked our projects on a scale of 1-3 since we were comparing 3 projects, with the greatest added total being the design that best met our requirements. We had a decision matrix for high priority and low priority criterion with high priority weighted at 1 and low priority weighted at 0.5. This was to ensure that our most important priorities are met first.

High Priority Matrix (weighted at 1)

Projects	Efficacy	Variety	Public Spaces	Durability	Weight and	Totals
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					Volume	
Jack's "Foam"	2	3	3	1	2	11
Keenan's "L"	1	1	3	2	1	8
Meredith/ Kayla's "Bolt"	3	2	3	3	3	14

Low Priority Matrix (weighted at 0.5)

Projects	Speed of Deployment	Cost of Materials	Total:
Jack's "Foam"	2	3	5
Keenan's "L"	3	1	4
Kayla/Meredith's "Bolt"	1	2	3

Calculated Total:

Jack's "Foam" = $11(1) + 5(0.5) = 13.5$

Keenan's "L" = $8(1) + 4(0.5) = 10$

Kayla/Meredith's "Bolt" = $14(1) + 3(0.5) = 15.5$

Thus, our chosen Design is Kayla/Meredith's "Bolt".

Plan of Action:

https://docs.google.com/spreadsheets/d/1_m3yd3sAysI9jugkCP8WFdhGFqTZg_dD300qPprW-NQ/edit?usp=sharing

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

- For a level 4, the plan of action should be detailed enough that someone could repeat the same steps and testing for effectiveness like a cookbook or recipe.

Blueprint:

- Describe all of your design's details such that someone unfamiliar with your project could build your prototype.
- Include sketches (KAYLA), dimensions(JACK) , CAD(KEENAN) (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

SKETCHES: